Command Prompt

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Microsoft Windows [Version 10.0.19045.6456]
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C:\Users\Student>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : rcoe.co.in
   Link-local IPv6 Address . . . . . : fe80::68dc:e693:5211:219e%8
   IPv4 Address. . . . . . . . . . . : 172.16.6.39
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 172.16.6.1

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : Dlink

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C:\Users\Student>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LP-74
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : rcoe.co.in

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : E4-02-9B-A9-2A-C8
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : E6-02-9B-A9-2A-C7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : rcoe.co.in
   Description . . . . . . . . . . . : Realtek PCIe GBE Family Controller
   Physical Address. . . . . . . . . : 1C-39-47-E8-E0-E3
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::68dc:e693:5211:219e%8(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.16.6.39(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 10 July 2026 09:25:34
   Lease Expires . . . . . . . . . . : 10 July 2026 15:58:23
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   Lease Obtained. . . . . . . . . . : 10 July 2026 09:25:34
   Lease Expires . . . . . . . . . . : 10 July 2026 15:58:23
   Default Gateway . . . . . . . . . : 172.16.6.1
   DHCP Server . . . . . . . . . . . : 172.16.6.1
   DHCPv6 IAID . . . . . . . . . . . : 152844615
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-1C-DF-D7-1C-39-47-E8-E0-E3
   DNS Servers . . . . . . . . . . . : 160.160.0.1
   Primary WINS Server . . . . . . . : 160.160.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : Dlink
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 3160
   Physical Address. . . . . . . . . : E4-02-9B-A9-2A-C7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

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C:\Users\Student>getmac

Physical Address    Transport Name
=================== ==========================================================
1C-39-47-E8-E0-E3   \Device\Tcpip_{60ECAF90-CB21-4073-9488-9C6F2B471332}
E4-02-9B-A9-2A-C7   Media disconnected

C:\Users\Student>hostname
LP-74

C:\Users\Student>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.6.39:7680       172.16.6.28:11055      TIME_WAIT
  TCP    172.16.6.39:7680       172.16.6.28:11129      TIME_WAIT
  TCP    172.16.6.39:7680       MEOW:52065             TIME_WAIT
  TCP    172.16.6.39:7680       MEOW:52070             TIME_WAIT
  TCP    172.16.6.39:7680       MEOW:52074             TIME_WAIT
  TCP    172.16.6.39:61718      lcdels-in-f188:5228    ESTABLISHED
  TCP    172.16.6.39:61939      a23-212-0-11:https     CLOSE_WAIT
  TCP    172.16.6.39:61948      40.104.68.50:https     ESTABLISHED
  TCP    172.16.6.39:61954      52.98.123.210:https    ESTABLISHED
  TCP    172.16.6.39:61958      52.98.59.18:https      ESTABLISHED
  TCP    172.16.6.39:62008      lcbome-in-f139:https   ESTABLISHED
  TCP    172.16.6.39:62013      fx-in-f113:https       TIME_WAIT
  TCP    172.16.6.39:62029      hkg07s51-in-f1:https   ESTABLISHED
  TCP    172.16.6.39:62039      pnbomb-bp-in-f14:https  TIME_WAIT
  TCP    172.16.6.39:62041      lcbomp-in-f101:https   TIME_WAIT
  TCP    172.16.6.39:62043      fx-in-f102:https       ESTABLISHED
  TCP    172.16.6.39:62044      fx-in-f94:https        ESTABLISHED
  TCP    172.16.6.39:62045      hkg12s09-in-f14:https  ESTABLISHED
  TCP    172.16.6.39:62049      104.208.16.90:https    TIME_WAIT
  TCP    172.16.6.39:62052      ly-in-f95:https        ESTABLISHED
  TCP    172.16.6.39:62053      ly-in-f95:https        ESTABLISHED
  TCP    172.16.6.39:62054      a23-212-0-11:https     SYN_SENT
  TCP    172.16.6.39:63785      4.213.25.240:https     ESTABLISHED

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C:\Users\Student>netstat/a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:80             LP-74:0                LISTENING
  TCP    0.0.0.0:135            LP-74:0                LISTENING
  TCP    0.0.0.0:445            LP-74:0                LISTENING
  TCP    0.0.0.0:1433           LP-74:0                LISTENING
  TCP    0.0.0.0:2383           LP-74:0                LISTENING
  TCP    0.0.0.0:5040           LP-74:0                LISTENING
  TCP    0.0.0.0:7680           LP-74:0                LISTENING
  TCP    0.0.0.0:49664          LP-74:0                LISTENING
  TCP    0.0.0.0:49665          LP-74:0                LISTENING
  TCP    0.0.0.0:49666          LP-74:0                LISTENING
  TCP    0.0.0.0:49667          LP-74:0                LISTENING
  TCP    0.0.0.0:49668          LP-74:0                LISTENING
  TCP    0.0.0.0:49670          LP-74:0                LISTENING
  TCP    0.0.0.0:49671          LP-74:0                LISTENING
  TCP    127.0.0.1:1434         LP-74:0                LISTENING
  TCP    172.16.6.39:139        LP-74:0                LISTENING
  TCP    172.16.6.39:7680       PC-307:50778           TIME_WAIT
  TCP    172.16.6.39:7680       PC-307:50788           TIME_WAIT
  TCP    172.16.6.39:7680       PC-307:50790           TIME_WAIT
  TCP    172.16.6.39:7680       PC-307:50800           TIME_WAIT
  TCP    172.16.6.39:7680       MEOW:52127             TIME_WAIT
  TCP    172.16.6.39:7680       MEOW:52178             TIME_WAIT
  TCP    172.16.6.39:61718      lcdels-in-f188:5228    ESTABLISHED
  TCP    172.16.6.39:61948      40.104.68.50:https     ESTABLISHED
  TCP    172.16.6.39:61954      52.98.123.210:https    ESTABLISHED
  TCP    172.16.6.39:61958      52.98.59.18:https      ESTABLISHED
  TCP    172.16.6.39:62008      lcbome-in-f139:https   ESTABLISHED
  TCP    172.16.6.39:62052      ly-in-f95:https        TIME_WAIT
  TCP    172.16.6.39:62053      ly-in-f95:https        ESTABLISHED
  TCP    172.16.6.39:62055      72.153.5.60:https      TIME_WAIT
  TCP    172.16.6.39:62057      72.153.5.129:https     TIME_WAIT
  TCP    172.16.6.39:62058      72.153.5.63:https      TIME_WAIT
  TCP    172.16.6.39:62059      72.153.5.134:https     TIME_WAIT
  TCP    172.16.6.39:62067      ly-in-f100:https       ESTABLISHED
  TCP    172.16.6.39:62068      ly-in-f100:https       ESTABLISHED
  TCP    172.16.6.39:62069      fx-in-f95:https        ESTABLISHED
  TCP    172.16.6.39:63785      4.213.25.240:https     ESTABLISHED
  TCP    [::]:80                LP-74:0                LISTENING
  TCP    [::]:135               LP-74:0                LISTENING
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  TCP    [::]:80                LP-74:0                LISTENING
  TCP    [::]:135               LP-74:0                LISTENING
  TCP    [::]:445               LP-74:0                LISTENING
  TCP    [::]:1433              LP-74:0                LISTENING
  TCP    [::]:2383              LP-74:0                LISTENING
  TCP    [::]:7680              LP-74:0                LISTENING
  TCP    [::]:49664             LP-74:0                LISTENING
  TCP    [::]:49665             LP-74:0                LISTENING
  TCP    [::]:49666             LP-74:0                LISTENING
  TCP    [::]:49667             LP-74:0                LISTENING
  TCP    [::]:49668             LP-74:0                LISTENING
  TCP    [::]:49670             LP-74:0                LISTENING
  TCP    [::]:49671             LP-74:0                LISTENING
  TCP    [::1]:1434             LP-74:0                LISTENING
  UDP    0.0.0.0:123            *:*
  UDP    0.0.0.0:500            *:*
  UDP    0.0.0.0:4500           *:*
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
  UDP    0.0.0.0:54163          *:*
  UDP    0.0.0.0:62422          *:*
  UDP    127.0.0.1:1900         *:*
  UDP    127.0.0.1:51071        *:*
  UDP    127.0.0.1:54180        *:*
  UDP    172.16.6.39:137        *:*
  UDP    172.16.6.39:138        *:*
  UDP    172.16.6.39:1900       *:*
  UDP    172.16.6.39:2177       *:*
  UDP    172.16.6.39:51070      *:*
  UDP    [::]:123               *:*
  UDP    [::]:500               *:*
  UDP    [::]:4500              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5355              *:*
  UDP    [::1]:1900             *:*
  UDP    [::1]:51069            *:*
  UDP    [fe80::68dc:e693:5211:219e%8]:1900  *:*
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  UDP    [::1]:51069              *:*
  UDP    [fe80::68dc:e693:5211:219e%8]:1900  *:*
  UDP    [fe80::68dc:e693:5211:219e%8]:2177  *:*
  UDP    [fe80::68dc:e693:5211:219e%8]:51068  *:*

C:\Users\Student>netstat/all

Displays protocol statistics and current TCP/IP network connections.

NETSTAT [-a] [-b] [-e] [-f] [-n] [-o] [-p proto] [-r] [-s] [-t] [-x] [-y] [interval]

  -a             Displays all connections and listening ports.
  -b             Displays the executable involved in creating each connection or
                 listening port. In some cases well-known executables host
                 multiple independent components, and in these cases the
                 sequence of components involved in creating the connection
                 or listening port is displayed. In this case the executable
                 name is in [] at the bottom, on top is the component it called,
                 and so forth until TCP/IP was reached. Note that this option
                 can be time-consuming and will fail unless you have sufficient
                 permissions.
  -e             Displays Ethernet statistics. This may be combined with the -s
                 option.
  -f             Displays Fully Qualified Domain Names (FQDN) for foreign
                 addresses.
  -n             Displays addresses and port numbers in numerical form.
  -o             Displays the owning process ID associated with each connection.
  -p proto       Shows connections for the protocol specified by proto; proto
                 may be any of: TCP, UDP, TCPv6, or UDPv6.  If used with the -s
                 option to display per-protocol statistics, proto may be any of:
                 IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, or UDPv6.
  -q             Displays all connections, listening ports, and bound
                 nonlistening TCP ports. Bound nonlistening ports may or may not
                 be associated with an active connection.
  -r             Displays the routing table.
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Select Command Prompt

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                 endpoints.
  -y             Displays the TCP connection template for all connections.
                 Cannot be combined with the other options.
  interval       Redisplays selected statistics, pausing interval seconds
                 between each display.  Press CTRL+C to stop redisplaying
                 statistics.  If omitted, netstat will print the current
                 configuration information once.


C:\Users\Student>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a             Displays current ARP entries by interrogating the current
                 protocol data.  If inet_addr is specified, the IP and Physical
                 addresses for only the specified computer are displayed.  If
                 more than one network interface uses ARP, entries for each ARP
                 table are displayed.
  -g             Same as -a.
  -v             Displays current ARP entries in verbose mode.  All invalid
                 entries and entries on the loop-back interface will be shown.
  inet_addr      Specifies an internet address.
  -N if_addr     Displays the ARP entries for the network interface specified
                 by if_addr.
  -d             Deletes the host specified by inet_addr. inet_addr may be
                 wildcarded with * to delete all hosts.
  -s             Adds the host and associates the Internet address inet_addr
                 with the Physical address eth_addr.  The Physical address is
                 given as 6 hexadecimal bytes separated by hyphens. The entry
                 is permanent.
  eth_addr       Specifies a physical address.
  if_addr        If present, this specifies the Internet address of the
                 interface whose address translation table should be modified.
                 If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

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C:\Users\Student>
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```
C:\Users\Student>arp/all

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a              Displays current ARP entries by interrogating the current
                  protocol data.  If inet_addr is specified, the IP and Physical
                  addresses for only the specified computer are displayed.  If
                  more than one network interface uses ARP, entries for each ARP
                  table are displayed.
  -g              Same as -a.
  -v              Displays current ARP entries in verbose mode.  All invalid
                  entries and entries on the loop-back interface will be shown.
  inet_addr       Specifies an internet address.
  -N if_addr      Displays the ARP entries for the network interface specified
                  by if_addr.
  -d              Deletes the host specified by inet_addr. inet_addr may be
                  wildcarded with * to delete all hosts.
  -s              Adds the host and associates the Internet address inet_addr
                  with the Physical address eth_addr.  The Physical address is
                  given as 6 hexadecimal bytes separated by hyphens. The entry
                  is permanent.
  eth_addr        Specifies a physical address.
  if_addr         If present, this specifies the Internet address of the
                  interface whose address translation table should be modified.
                  If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

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C:\Users\Student>arp -a

Interface: 172.16.6.39 --- 0x8
  Internet Address      Physical Address      Type
  172.16.6.1            b0-b1-a1-b5-f7-6f     dynamic
  172.16.6.15           40-8d-5c-a5-26-db     dynamic
  172.16.6.17           40-8d-5c-a5-26-d7     dynamic
  172.16.6.18           40-8d-5c-a5-26-da     dynamic
  172.16.6.20           40-8d-5c-a5-27-b9     dynamic
  172.16.6.28           1c-39-47-28-00-9c     dynamic
  172.16.6.30           1c-39-47-28-01-6a     dynamic
  172.16.6.31           1c-39-47-41-33-37     dynamic
  172.16.6.33           1c-39-47-e6-50-9c     dynamic
  172.16.6.34           1c-39-47-e6-50-66     dynamic
  172.16.6.37           1c-39-47-e6-4f-f3     dynamic
  172.16.6.40           1c-39-47-e6-50-54     dynamic
  172.16.6.255          ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\Student>
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C:\Users\Student>tracert google.co.in

Tracing route to google.co.in [192.178.193.94]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.16.6.1
  2     *        *        *     Request timed out.
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     2 ms     2 ms     3 ms  142.251.225.29
  9     5 ms     3 ms     3 ms  192.178.110.208
 10     3 ms     2 ms     2 ms  142.251.78.191
 11     3 ms     2 ms     2 ms  192.178.45.10
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        2 ms  ly-in-f94.1e100.net [192.178.193.94]

Trace complete.
```